FINAL PROJECT REPORT

Group 2
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Introduction

- An ongoing trend of “loneliness epidemic”
- Research question: How has the average daily time spent alone changed over time among different age groups
- Analyzing this trend through social data can reveal generational shifts, identify at-risk groups, and provide insight into broader societal changes affecting well-being.

Dataset Overview

Data Source: *Our World in Data,* based on U.S. Bureau of Labor Statistics (2023)

- Reliable and free
- Covers many data categories
- Regularly updated

Why this dataset and how it connects to our research question?

1. Covers average daily time spent alone and with others (2003-2023)
2. Includes key variable “time spent alone” by age group
3. Provides a basis for analyzing social isolation risks



Dataset Structure & Key Variables

- Format: CSV tabular data
- Time Span: 2003 to 2023 (time series)
- Size: 105 rows × 8 columns

	Entity	Code	Year	Time spent with other people over time - All people - Alone	Time spent with other people over time - All people - Friend	Time spent with other people over time - All people - Children	Time spent with other people over time - All people - Family	Time spent with other people over time - All people - Partner	Time spent with other people over time - All people - Co-worker
0	15-29 years	NaN	2003	3.831570	2.209241	1.676127	2.331077	1.283852	0.314280
1	15-29 years	NaN	2004	3.856597	2.066119	1.727781	2.284257	1.336806	0.357941
2	15-29 years	NaN	2005	3.591629	2.068284	1.942361	2.358772	1.357930	0.308502
3	15-29 years	NaN	2006	3.400003	1.764779	1.854058	2.428823	1.268695	0.261143
4	15-29 years	NaN	2007	3.669142	1.691049	1.797517	2.359614	1.246309	0.278988



Key Variables

- Entity: Age group (e.g., 15-29 years, 30-44 years)
- Year: 2003-2023
- Average daily time (in hours) spent with different types of people: alone, with friends, children, family, partners, and co-workers.

Limitations or challenges

Overlapping Relationship Categories

One person may belong to multiple categories, which can lead to double counting.

Missing Metadata

Some fields like region codes or activity contexts are not included.



Exclusion of Non-social Activities

Time spent on work, sleep, and personal care is not recorded.

Methodology

Data Visualizations

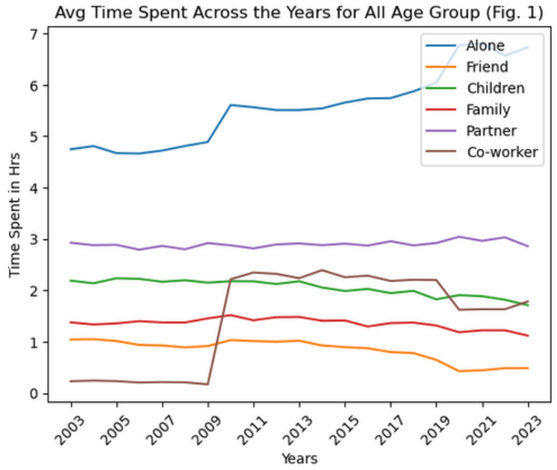
We used line charts and an overlapping bar chart to explore trends in time spent alone across different age groups.

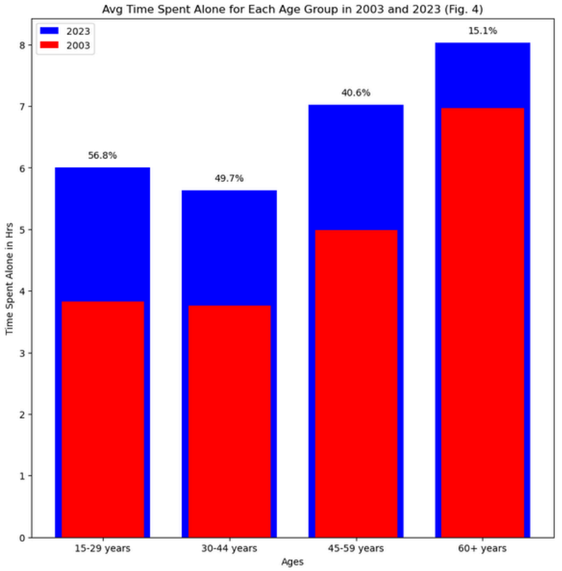
Line Charts

Ideal for showing changes over time, line charts allowed us to compare trends across multiple age groups and observe long-term patterns in alone time.

Overlapping Bar Chart

This helped us compare changes between two specific time points, making it easy to visualize which age groups experienced the most significant increases in alone time.

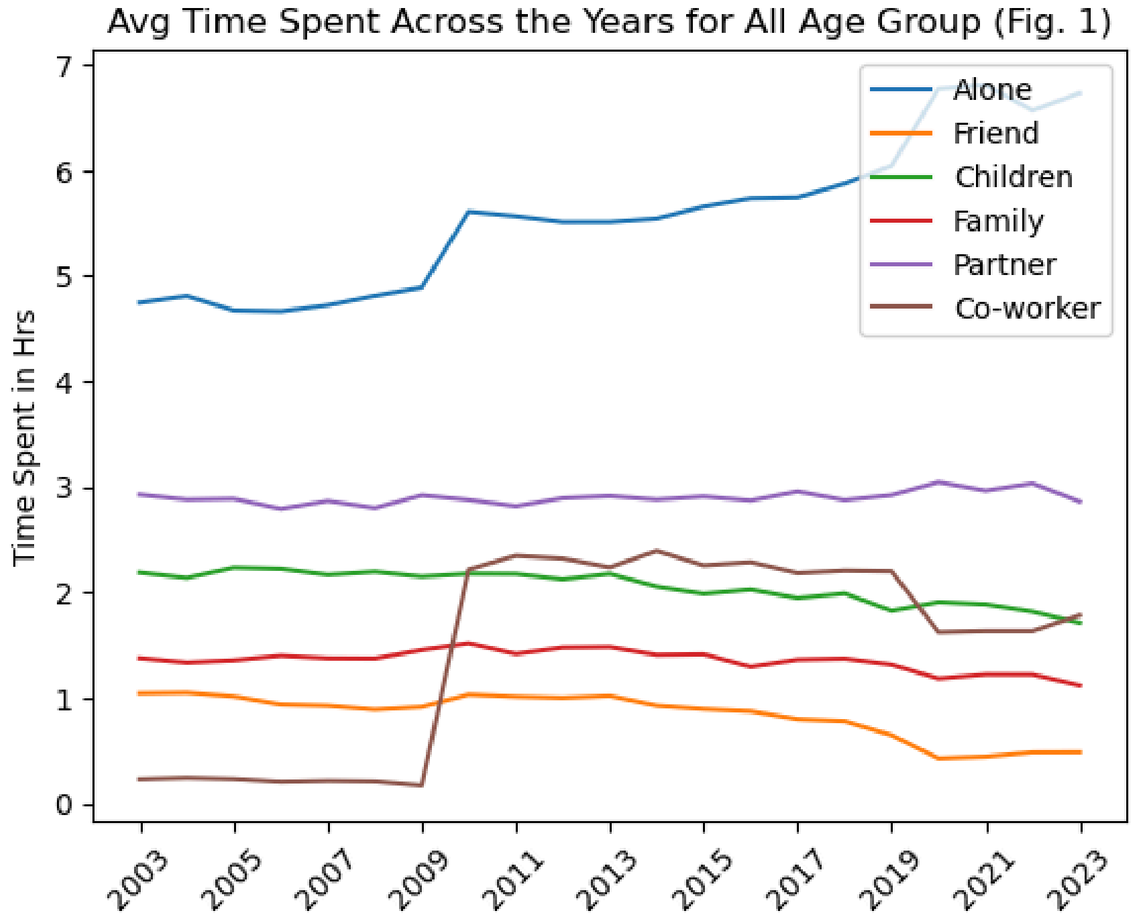






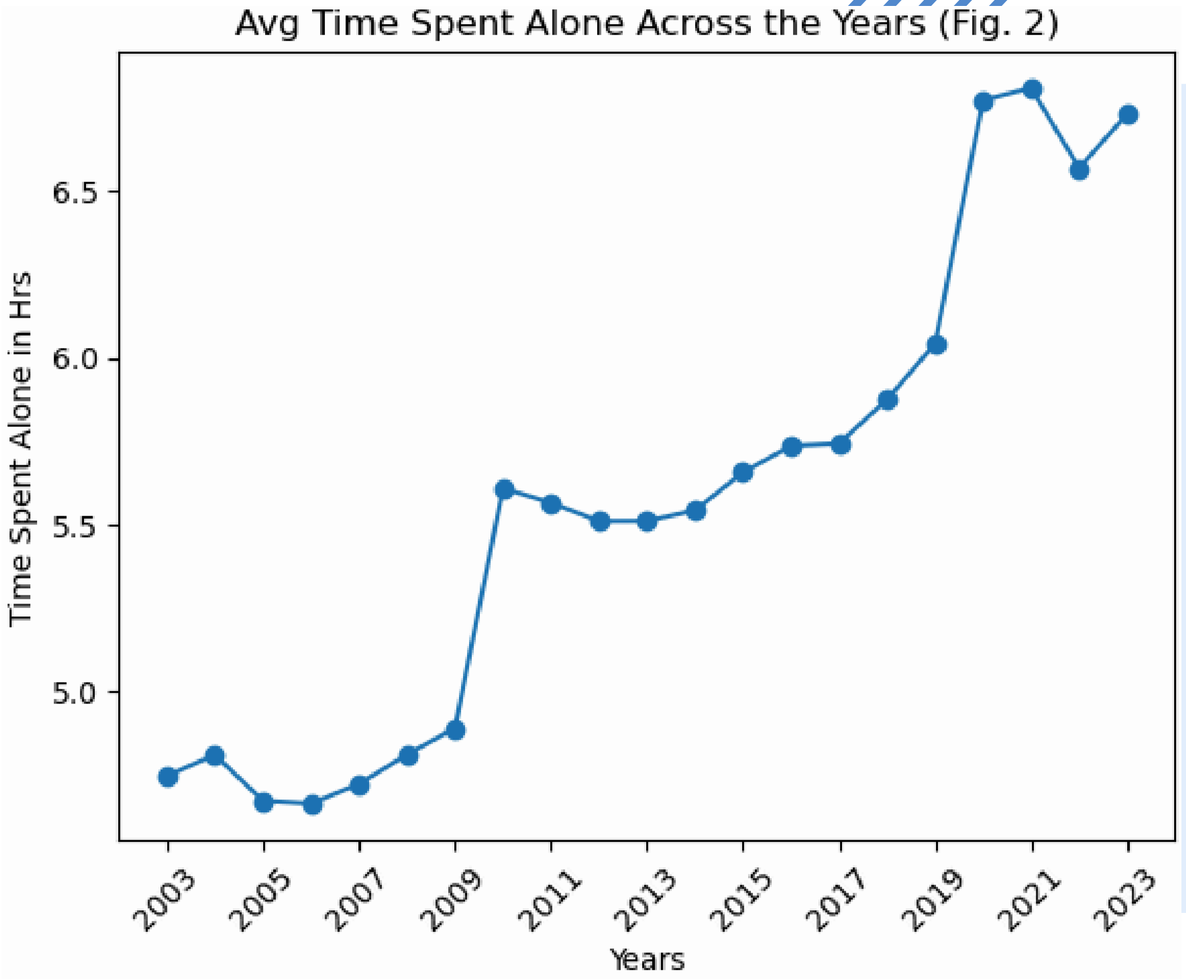
Results and Discussion

- **Increase of time spent alone over the years.**
- **Decline and stagnation of other categories of time spent with other people.**
- **‘Co-Worker’ is an exception, with a huge increase in 2009, leading it to have a large growth since 2003.**



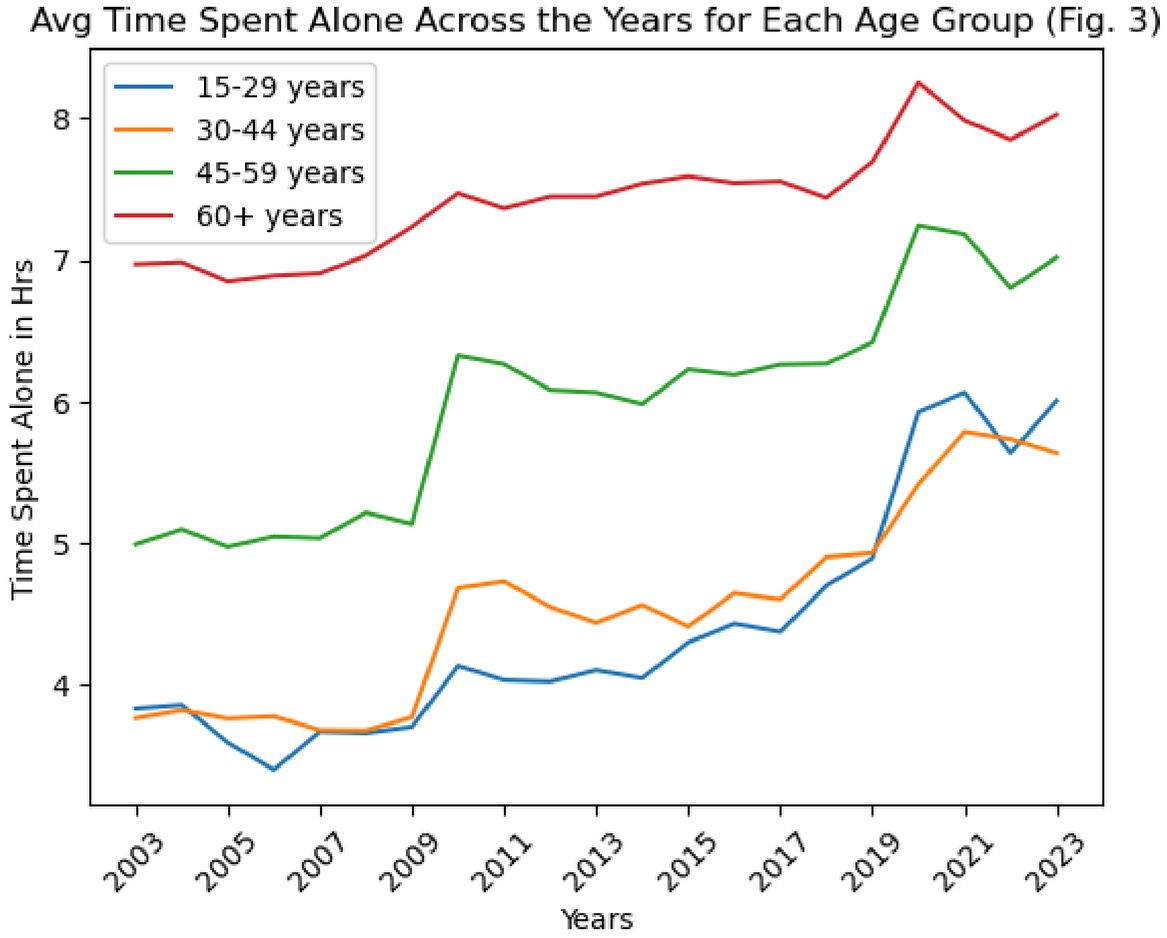
Results and Discussion

- **Surge in the time spent alone between 2009-2010 and 2019-2020.**
- **A 2 hour increase in time spent alone over 20 years.**



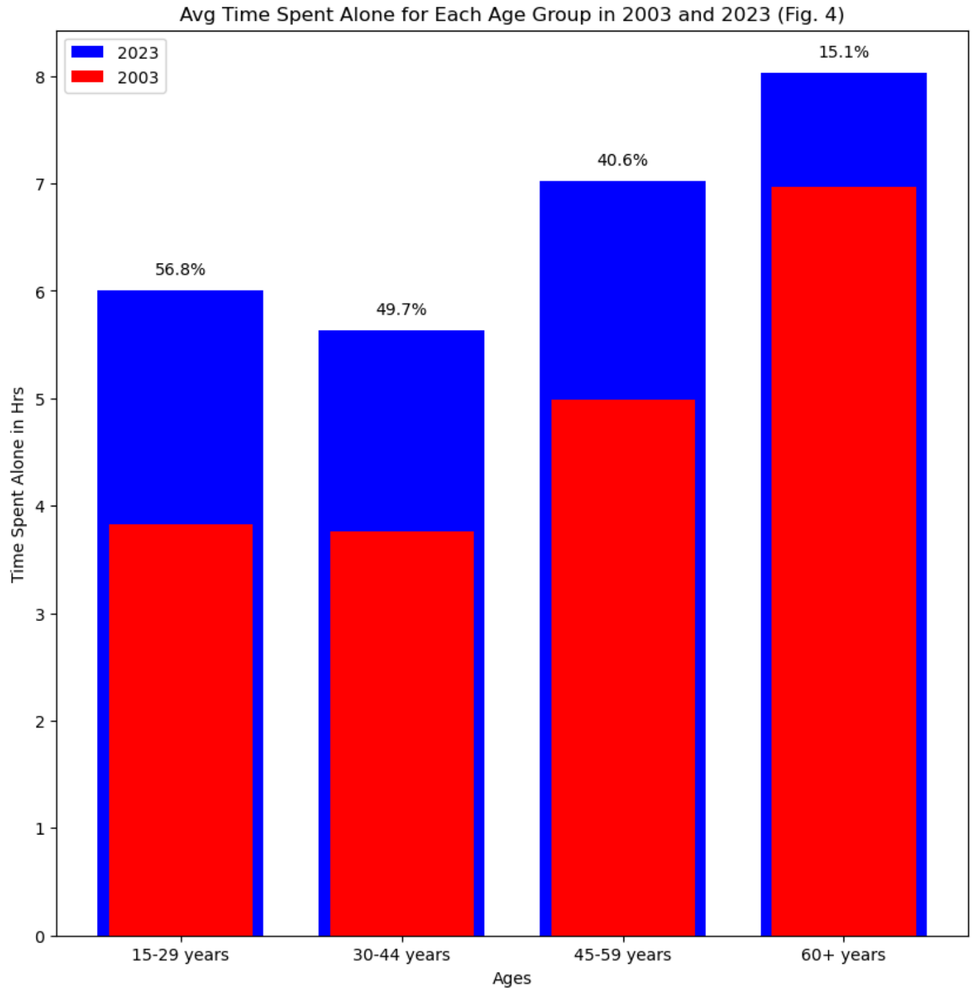
Results and Discussion

- **60+ age group is constantly spending the most time alone.**
- **15-29 age group spent more time alone in 2023 than the 30-44 age group.**
- **Trend of youngest spending least time alone is not always true.**



Results and Discussion

- 60+ age group had lowest increase of time spent alone between 2003 and 2023 with a 15.1% increase.
- 15-29 age group had greatest increase of time spent alone between 2003 and 2023 with a 56.8% increase.
- Indicates that younger age groups have a higher increase in time spent alone over the years.



Conclusion

- Over the past 20 years, Americans have spent increasingly more time alone
- Sharp rise during financial crisis & COVID-19
- The government or policymakers can better develop social programs or community support
- Future research: explore causes (tech, lifestyle, mental health)



References

Our World in Data. (2023). Who Americans spend their time with, by age. https://ourworldindata.org/grapher/time-spent-with-relationships-by-age-us

Ross, E. M. (2024, October 25). What is causing our epidemic of loneliness and how can we fix it?. Harvard Graduate School of Education. https://www.gse.harvard.edu/ideas/usable-knowledge/24/10/what-causing-our-epidemic-loneliness-and-how-can-we-fix-it

THANK YOU!